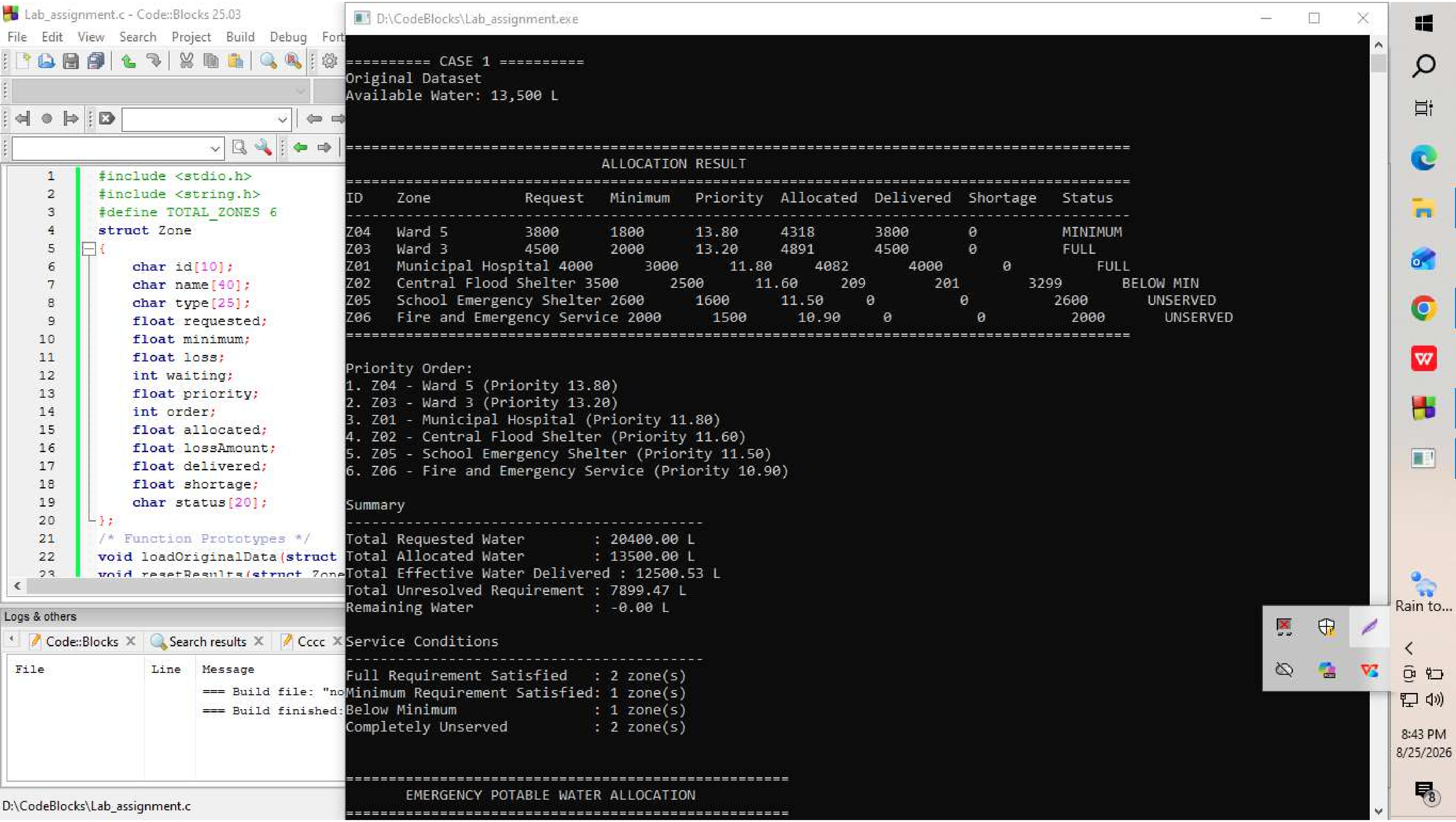
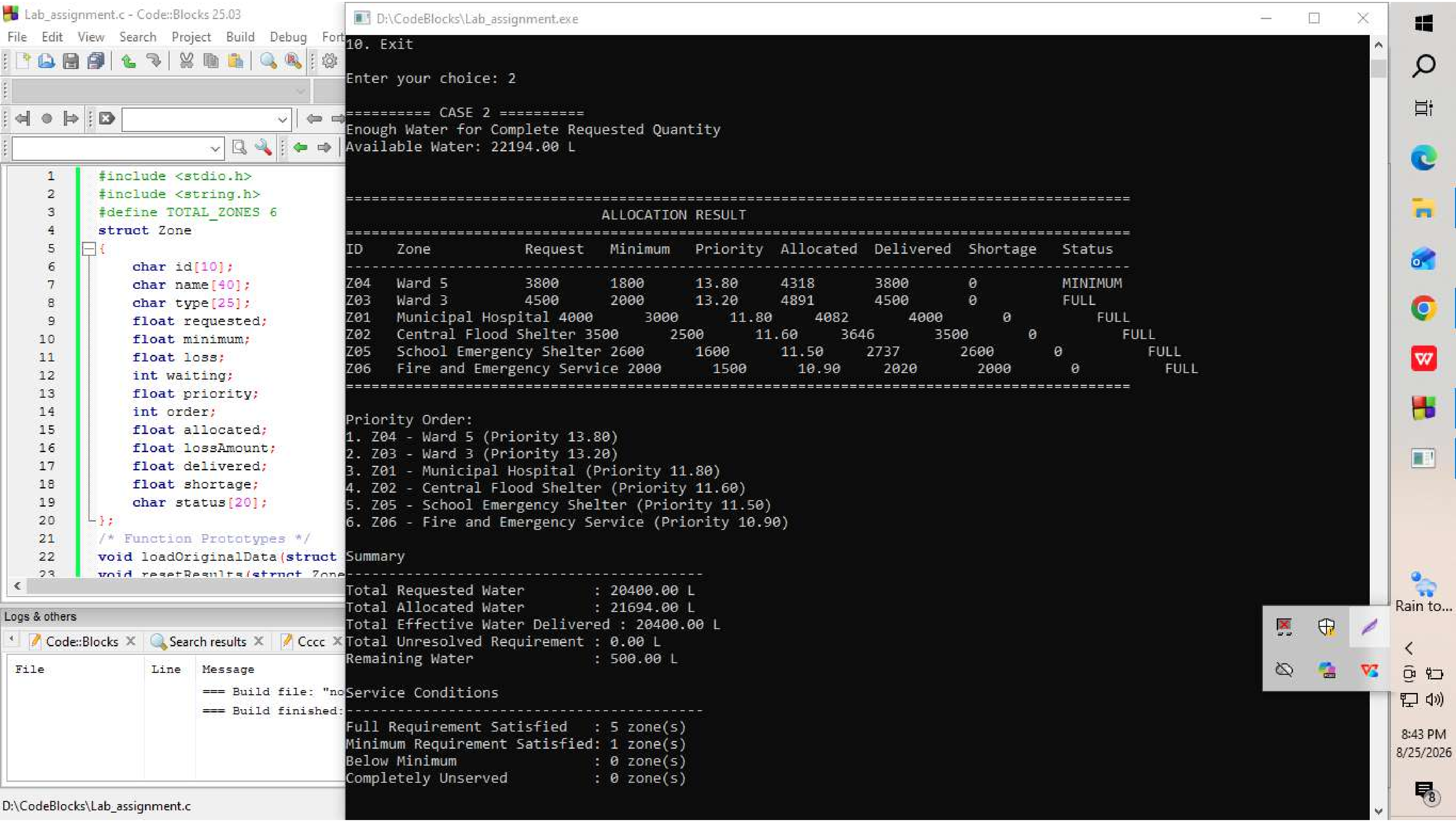

Lab_assignment.c - Code::Blocks 25.03
File Edit View Search Project Build Debug
#include <stdio.h>
#include <string.h>
#define TOTAL_ZONES 6
struct Zone
{
char id[10];
char name[40];
char type[25];
float requested;
float minimum;
float loss;
int waiting;
float priority;
int order;
float allocated;
float lossAmount;
float delivered;
float shortage;
char status[20];
};
/* Function Prototypes */
void loadOriginalData(struct
Logs & others
Code::Blocks
Search results
Cccc
File
Line
Message
=== Build file: "no
=== Build finished:
D:\CodeBlocks\Lab_assignment.c
D:\CodeBlocks\Lab_assignment.exe
========== CASE 1 ==========
Original Dataset
Available Water: 13,500 L
==
ALLOCATION RESULT
==
ID Zone Request Minimum Priority Allocated Delivered Shortage Status
----
Z04 Ward 5 3800 1800 13.80 4318 3800 0 MINIMUM
Z03 Ward 3 4500 2000 13.20 4891 4500 0 FULL
Z01 Municipal Hospital 4000 3000 11.80 4082 4000 0 FULL
Z02 Central Flood Shelter 3500 2500 11.60 209 201 3299 BELOW MIN
Z05 School Emergency Shelter 2600 1600 11.50 0 0 2600 UNSERVED
Z06 Fire and Emergency Service 2000 1500 10.90 0 0 2000 UNSERVED
==
Priority Order:
1. Z04 - Ward 5 (Priority 13.80)
2. Z03 - Ward 3 (Priority 13.20)
3. Z01 - Municipal Hospital (Priority 11.80)
4. Z02 - Central Flood Shelter (Priority 11.60)
5. Z05 - School Emergency Shelter (Priority 11.50)
6. Z06 - Fire and Emergency Service (Priority 10.90)
Summary

Total Requested Water : 20400.00 L
Total Allocated Water : 13500.00 L
Total Effective Water Delivered : 12500.53 L
Total Unresolved Requirement : 7899.47 L
Remaining Water : -0.00 L
Service Conditions

Full Requirement Satisfied : 2 zone(s)
Minimum Requirement Satisfied: 1 zone(s)
Below Minimum : 1 zone(s)
Completely Unserved : 2 zone(s)
==
EMERGENCY POTABLE WATER ALLOCATION
==
Rain to...
8:43 PM
8/25/2026


Lab_assignment.c - Code::Blocks 25.03
File Edit View Search Project Build Debug
#include <stdio.h>
#include <string.h>
#define TOTAL_ZONES 6
struct Zone
{
char id[10];
char name[40];
char type[25];
float requested;
float minimum;
float loss;
int waiting;
float priority;
int order;
float allocated;
float lossAmount;
float delivered;
float shortage;
char status[20];
};
/* Function Prototypes */
void loadOriginalData(struct
Logs & others
Code::Blocks
Search results
Cccc
File
Line
Message
=== Build file: "no
=== Build finished:
D:\CodeBlocks\Lab_assignment.c
D:\CodeBlocks\Lab_assignment.exe
10. Exit
Enter your choice: 2
========== CASE 2 ==========
Enough Water for Complete Requested Quantity
Available Water: 22194.00 L
ALLOCATION RESULT
ID Zone Request Minimum Priority Allocated Delivered Shortage Status
Z04 Ward 5 3800 1800 13.80 4318 3800 0 MINIMUM
Z03 Ward 3 4500 2000 13.20 4891 4500 0 FULL
Z01 Municipal Hospital 4000 3000 11.80 4082 4000 0 FULL
Z02 Central Flood Shelter 3500 2500 11.60 3646 3500 0 FULL
Z05 School Emergency Shelter 2600 1600 11.50 2737 2600 0 FULL
Z06 Fire and Emergency Service 2000 1500 10.90 2020 2000 0 FULL
Priority Order:
1. Z04 - Ward 5 (Priority 13.80)
2. Z03 - Ward 3 (Priority 13.20)
3. Z01 - Municipal Hospital (Priority 11.80)
4. Z02 - Central Flood Shelter (Priority 11.60)
5. Z05 - School Emergency Shelter (Priority 11.50)
6. Z06 - Fire and Emergency Service (Priority 10.90)
Summary
Total Requested Water : 20400.00 L
Total Allocated Water : 21694.00 L
Total Effective Water Delivered : 20400.00 L
Total Unresolved Requirement : 0.00 L
Remaining Water : 500.00 L
Service Conditions
Full Requirement Satisfied : 5 zone(s)
Minimum Requirement Satisfied: 1 zone(s)
Below Minimum : 0 zone(s)
Completely Unserved : 0 zone(s)
Rain to...
8:43 PM
8/25/2026

Lab_assignment.c - Code::Blocks 25.03

File Edit View Search Project Build Debug Fort

```c
#include <stdio.h>
#include <string.h>
#define TOTAL_ZONES 6
struct Zone
{
    char id[10];
    char name[40];
    char type[25];
    float requested;
    float minimum;
    float loss;
    int waiting;
    float priority;
    int order;
    float allocated;
    float lossAmount;
    float delivered;
    float shortage;
    char status[20];
};
/* Function Prototypes */
void loadOriginalData(struct
void resetResults(struct Zone
```

Logs & others

Code::Blocks | Search results | Cccc

```
File        Line    Message
                    === Build file: "no
                    === Build finished:
```

D:\CodeBlocks\Lab_assignment.c

D:\CodeBlocks\Lab_assignment.exe

```
9. Find Highest Unresolved Requirement
10. Exit

Enter your choice: 3

========== CASE 3 ==========
Enter available water in litres: 8000


============================================================================================
                              ALLOCATION RESULT
============================================================================================
ID    Zone           Request   Minimum   Priority  Allocated  Delivered  Shortage   Status
--------------------------------------------------------------------------------------------
Z04   Ward 5         3800      1800      13.80     4318       3800       0          MINIMUM
Z03   Ward 3         4500      2000      13.20     2174       2000       2500       MINIMUM
Z01   Municipal Hospital 4000      3000      11.80     1508       1478       2522       BELOW MIN
Z02   Central Flood Shelter 3500      2500      11.60     0          0          3500       UNSERVED
Z05   School Emergency Shelter 2600      1600      11.50     0          0          2600       UNSERVED
Z06   Fire and Emergency Service 2000      1500      10.90     0          0          2000       UNSERVED
============================================================================================

Priority Order:
1. Z04 - Ward 5 (Priority 13.80)
2. Z03 - Ward 3 (Priority 13.20)
3. Z01 - Municipal Hospital (Priority 11.80)
4. Z02 - Central Flood Shelter (Priority 11.60)
5. Z05 - School Emergency Shelter (Priority 11.50)
6. Z06 - Fire and Emergency Service (Priority 10.90)

Summary
------------------------------------------
Total Requested Water        : 20400.00 L
Total Allocated Water        : 8000.00 L
Total Effective Water Delivered : 7277.75 L
Total Unresolved Requirement : 13122.25 L
Remaining Water              : 0.00 L

Service Conditions
------------------------------------------
Full Requirement Satisfied   : 0 zone(s)
Minimum Requirement Satisfied: 2 zone(s)
Below Minimum                : 1 zone(s)
Completely Unserved          : 3 zone(s)
```

Rain to...

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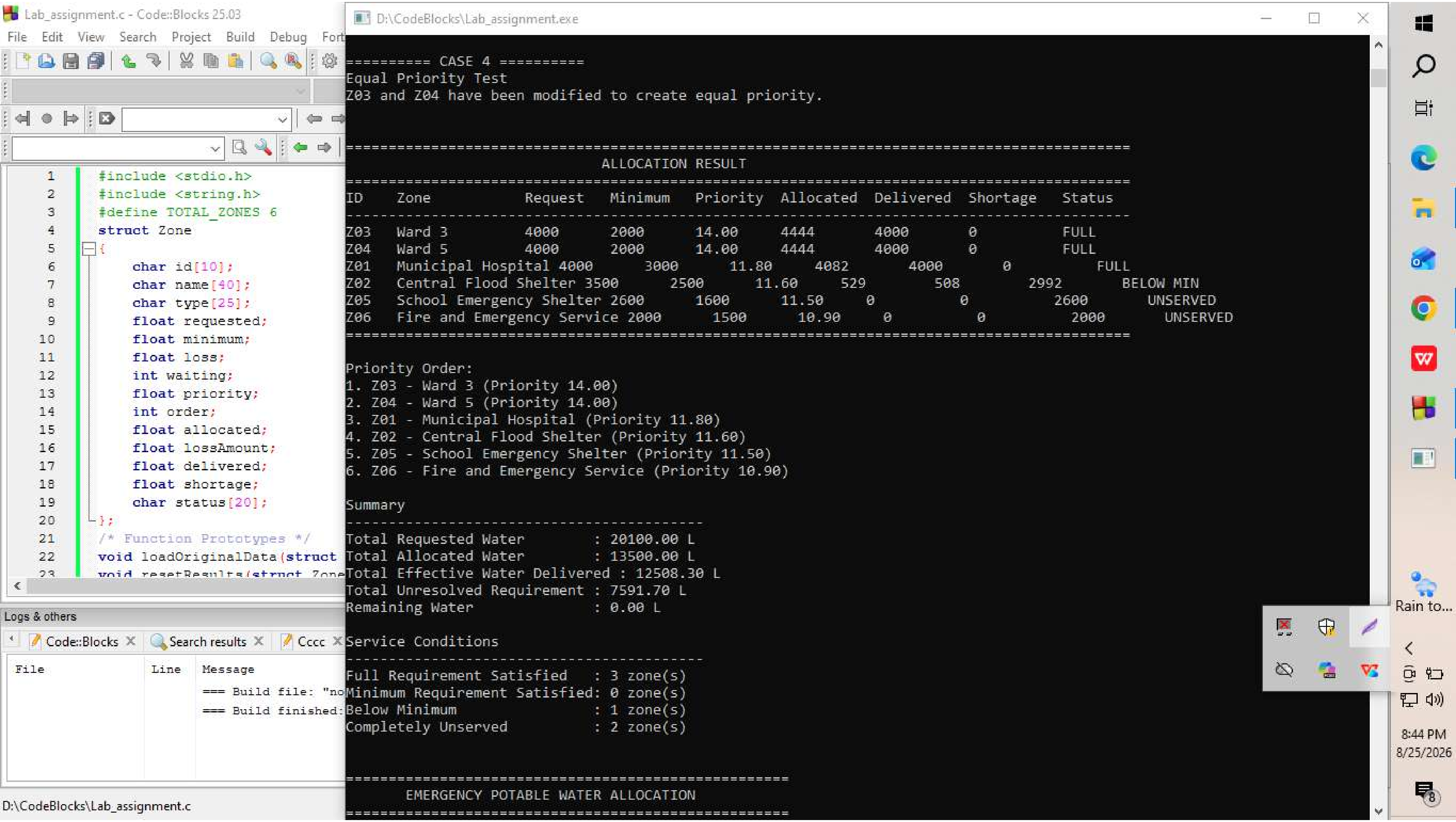
8/25/2026



Lab_assignment.c - Code::Blocks 25.03
File Edit View Search Project Build Debug
#include <stdio.h>
#include <string.h>
#define TOTAL_ZONES 6
struct Zone
{
char id[10];
char name[40];
char type[25];
float requested;
float minimum;
float loss;
int waiting;
float priority;
int order;
float allocated;
float lossAmount;
float delivered;
float shortage;
char status[20];
};
/* Function Prototypes */
void loadOriginalData(struct
Logs & others
Code::Blocks
Search results
Cccc
File
Line
Message
=== Build file: "no
=== Build finished:
D:\CodeBlocks\Lab_assignment.c
D:\CodeBlocks\Lab_assignment.exe
========== CASE 4 ==========
Equal Priority Test
Z03 and Z04 have been modified to create equal priority.
ALLOCATION RESULT
ID Zone Request Minimum Priority Allocated Delivered Shortage Status
Z03 Ward 3 4000 2000 14.00 4444 4000 0 FULL
Z04 Ward 5 4000 2000 14.00 4444 4000 0 FULL
Z01 Municipal Hospital 4000 3000 11.80 4082 4000 0 FULL
Z02 Central Flood Shelter 3500 2500 11.60 529 508 2992 BELOW MIN
Z05 School Emergency Shelter 2600 1600 11.50 0 0 2600 UNSERVED
Z06 Fire and Emergency Service 2000 1500 10.90 0 0 2000 UNSERVED
Priority Order:
1. Z03 - Ward 3 (Priority 14.00)
2. Z04 - Ward 5 (Priority 14.00)
3. Z01 - Municipal Hospital (Priority 11.80)
4. Z02 - Central Flood Shelter (Priority 11.60)
5. Z05 - School Emergency Shelter (Priority 11.50)
6. Z06 - Fire and Emergency Service (Priority 10.90)
Summary
Total Requested Water : 20100.00 L
Total Allocated Water : 13500.00 L
Total Effective Water Delivered : 12508.30 L
Total Unresolved Requirement : 7591.70 L
Remaining Water : 0.00 L
Service Conditions
Full Requirement Satisfied : 3 zone(s)
Minimum Requirement Satisfied: 0 zone(s)
Below Minimum : 1 zone(s)
Completely Unserved : 2 zone(s)
EMERGENCY POTABLE WATER ALLOCATION
Rain to...
8:44 PM
8/25/2026

Lab_assignment.c - Code::Blocks 25.03

File Edit View Search Project Build Debug Fort

```c
#include <stdio.h>
#include <string.h>
#define TOTAL_ZONES 6
struct Zone
{
    char id[10];
    char name[40];
    char type[25];
    float requested;
    float minimum;
    float loss;
    int waiting;
    float priority;
    int order;
    float allocated;
    float lossAmount;
    float delivered;
    float shortage;
    char status[20];
};
/* Function Prototypes */
void loadOriginalData(struct
void resetResults(struct Zone
```

Logs & others

Code::Blocks | Search results | Cccc

File	Line	Message
		=== Build file: "no
		=== Build finished:

D:\CodeBlocks\Lab_assignment.c

D:\CodeBlocks\Lab_assignment.exe

```
Enter your choice: 5

========== CASE 5 ==========
Residential zones are: Z03 and Z04
Enter residential Zone ID: Z03
Current distribution loss: 8.00%
Enter new distribution loss (%): 4

New loss of Z03 = 4.00%


=============================================================================================
                              ALLOCATION RESULT
=============================================================================================
ID    Zone           Request   Minimum   Priority  Allocated  Delivered  Shortage   Status
---------------------------------------------------------------------------------------------
Z04   Ward 5         3800      1800      13.80     4318       3800       0          MINIMUM
Z03   Ward 3         4500      2000      13.60     4688       4500       0          FULL
Z01   Municipal Hospital 4000      3000      11.80     4082       4000       0          FULL
Z02   Central Flood Shelter 3500      2500      11.60     413        396        3104       BELOW MIN
Z05   School Emergency Shelter 2600      1600      11.50     0          0          2600       UNSERVED
Z06   Fire and Emergency Service 2000      1500      10.90     0          0          2000       UNSERVED
=============================================================================================

Priority Order:
1. Z04 - Ward 5 (Priority 13.80)
2. Z03 - Ward 3 (Priority 13.60)
3. Z01 - Municipal Hospital (Priority 11.80)
4. Z02 - Central Flood Shelter (Priority 11.60)
5. Z05 - School Emergency Shelter (Priority 11.50)
6. Z06 - Fire and Emergency Service (Priority 10.90)

Summary
------------------------------------------
Total Requested Water        : 20400.00 L
Total Allocated Water        : 13500.00 L
Total Effective Water Delivered : 12696.18 L
Total Unresolved Requirement : 7703.82 L
Remaining Water              : 0.00 L

Service Conditions
------------------------------------------
Full Requirement Satisfied   : 2 zone(s)
Minimum Requirement Satisfied: 1 zone(s)
Below Minimum                : 1 zone(s)
```

Rain to...

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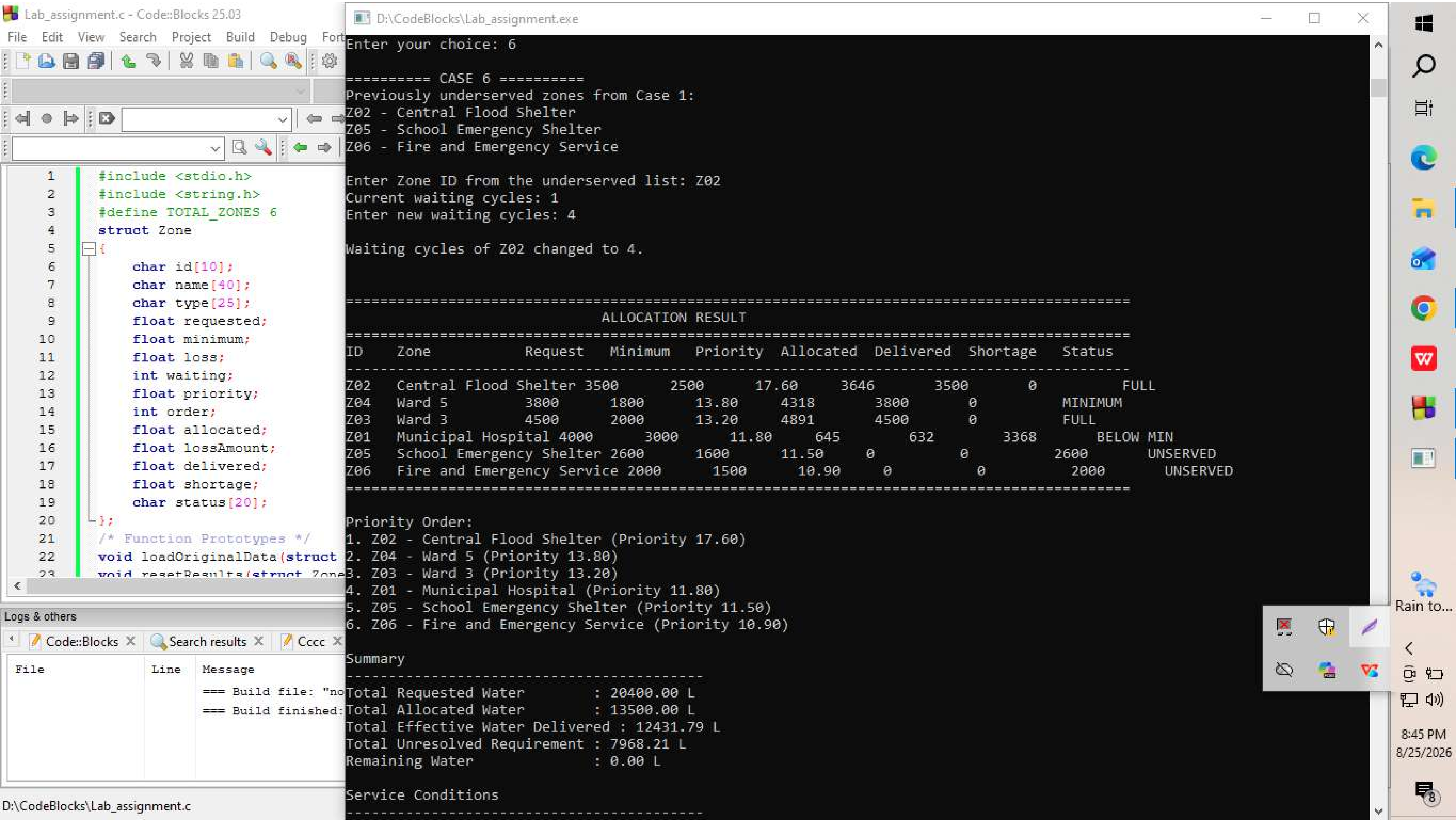
8/25/2026


Lab_assignment.c - Code::Blocks 25.03
File Edit View Search Project Build Debug Fort
#include <stdio.h>
#include <string.h>
#define TOTAL_ZONES 6
struct Zone
{
char id[10];
char name[40];
char type[25];
float requested;
float minimum;
float loss;
int waiting;
float priority;
int order;
float allocated;
float lossAmount;
float delivered;
float shortage;
char status[20];
};
/* Function Prototypes */
void loadOriginalData(struct
Logs & others
Code::Blocks
Search results
Cccc
File
Line
Message
=== Build file: "no
=== Build finished:
D:\CodeBlocks\Lab_assignment.c
D:\CodeBlocks\Lab_assignment.exe
Enter your choice: 6
========== CASE 6 ==========
Previously underserved zones from Case 1:
Z02 - Central Flood Shelter
Z05 - School Emergency Shelter
Z06 - Fire and Emergency Service
Enter Zone ID from the underserved list: Z02
Current waiting cycles: 1
Enter new waiting cycles: 4
Waiting cycles of Z02 changed to 4.
ALLOCATION RESULT
ID Zone Request Minimum Priority Allocated Delivered Shortage Status
Z02 Central Flood Shelter 3500 2500 17.60 3646 3500 0 FULL
Z04 Ward 5 3800 1800 13.80 4318 3800 0 MINIMUM
Z03 Ward 3 4500 2000 13.20 4891 4500 0 FULL
Z01 Municipal Hospital 4000 3000 11.80 645 632 3368 BELOW MIN
Z05 School Emergency Shelter 2600 1600 11.50 0 0 2600 UNSERVED
Z06 Fire and Emergency Service 2000 1500 10.90 0 0 2000 UNSERVED
Priority Order:
1. Z02 - Central Flood Shelter (Priority 17.60)
2. Z04 - Ward 5 (Priority 13.80)
3. Z03 - Ward 3 (Priority 13.20)
4. Z01 - Municipal Hospital (Priority 11.80)
5. Z05 - School Emergency Shelter (Priority 11.50)
6. Z06 - Fire and Emergency Service (Priority 10.90)
Summary
Total Requested Water : 20400.00 L
Total Allocated Water : 13500.00 L
Total Effective Water Delivered : 12431.79 L
Total Unresolved Requirement : 7968.21 L
Remaining Water : 0.00 L
Service Conditions
Rain to...
8:45 PM
8/25/2026

Lab_assignment.c - Code::Blocks 25.03

File Edit View Search Project Build Debug Fort

```c
#include <stdio.h>
#include <string.h>
#define TOTAL_ZONES 6
struct Zone
{
    char id[10];
    char name[40];
    char type[25];
    float requested;
    float minimum;
    float loss;
    int waiting;
    float priority;
    int order;
    float allocated;
    float lossAmount;
    float delivered;
    float shortage;
    char status[20];
};
/* Function Prototypes */
void loadOriginalData(struct
void resetResults(struct Zone
```

Logs & others

Code::Blocks | Search results | Cccc

File	Line	Message
		=== Build file: "no
		=== Build finished:

D:\CodeBlocks\Lab_assignment.c

D:\CodeBlocks\Lab_assignment.exe

```
10. Exit

Enter your choice: 7

========== CASE 7 ==========
Independent Test Case
Enter any available water quantity: 5000


==========================================================================================
                              ALLOCATION RESULT
==========================================================================================
ID    Zone           Request   Minimum   Priority  Allocated  Delivered  Shortage   Status
------------------------------------------------------------------------------------------
Z04   Ward 5         3800      1800      13.80     4318       3800       0          MINIMUM
Z03   Ward 3         4500      2000      13.20     682        627        3873       BELOW MIN
Z01   Municipal Hospital 4000      3000      11.80     0          0          4000       UNSERVED
Z02   Central Flood Shelter 3500      2500      11.60     0          0          3500       UNSERVED
Z05   School Emergency Shelter 2600      1600      11.50     0          0          2600       UNSERVED
Z06   Fire and Emergency Service 2000      1500      10.90     0          0          2000       UNSERVED
==========================================================================================

Priority Order:
1. Z04 - Ward 5 (Priority 13.80)
2. Z03 - Ward 3 (Priority 13.20)
3. Z01 - Municipal Hospital (Priority 11.80)
4. Z02 - Central Flood Shelter (Priority 11.60)
5. Z05 - School Emergency Shelter (Priority 11.50)
6. Z06 - Fire and Emergency Service (Priority 10.90)

Summary
----------------------------------------
Total Requested Water        : 20400.00 L
Total Allocated Water        : 5000.00 L
Total Effective Water Delivered : 4427.27 L
Total Unresolved Requirement : 15972.73 L
Remaining Water              : 0.00 L

Service Conditions
----------------------------------------
Full Requirement Satisfied   : 0 zone(s)
Minimum Requirement Satisfied: 1 zone(s)
Below Minimum                : 1 zone(s)
Completely Unserved          : 4 zone(s)
```

Rain to...

8:46 PM
8/25/2026